

Nishant Sharma

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About

AI Engineer and Researcher - specialist in Python Development, Agentic AI and MLOps Cloud Engineering. 5 years of experience with Python, 3 years of experience with Cloud and Systems Engineering and 1 year of experience with AI Agents and RAG systems. Available for Full time roles starting May 2026.

Technical Skills

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, Bash/Shell
- **Core Skills:** Data Structures, Algorithms, OOP, System Design (Microservices, Distributed Systems), DBMS, Agile/Scrum
- **Web and Backend:** Next.js, Node.js, Express, FastAPI, Django, Flask, Spring Boot, REST APIs, gRPC, GraphQL
- **Cloud and DevOps:** AWS (EC2, S3, RDS, IAM), Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, Prometheus, Grafana
- **AI and Research:** PyTorch, TensorFlow, LLMs (RAG, Fine-Tuning), LangGraph, Vector Databases (Pinecone/Milvus), MLOps (MLflow)
- **Databases and Storage:** PostgreSQL, MongoDB, Redis, DynamoDB

Education

- **New York University Tandon School of Engineering** | New York, US | Sep 2024 - Expected May 2026
Master of Science in Computer Engineering | GPA: 3.67
- **Dr. A.P.J. Abdul Kalam Technical University** | Delhi, India | Aug 2020 - Jun 2024
Bachelor of Technology in Computer Science & Engineering | 1st Division with Distinction

Work Experience

Machine Learning Teaching Assistant | NYU Tandon School of Engineering | NY, USA | Sep 2025 - Present

- Helped students understand Machine Learning application using **Pytorch, Numpy, Pandas, Matplotlib**, etc.
- Held weekly office hours to clear students' doubts in niche ML topics, and issues with code for assignments and project.
- Created **YouTube videos** explaining common doubts in Machine Learning.

Co-Founder | Ingelt Board | Delhi, India | Dec 2022 - Jul 2024

- **Architected a multi-tenant platform** supporting workflows for Teachers, Students, and Management, featuring real-time capabilities: **Group Chat** and **Live Streaming**.
- **Designed a cloud-native backend on AWS**, utilizing **S3** for scalable content storage mapped via **RDS**, and ensuring high availability through region-specific **EC2** instance management.
- **Implemented secure infrastructure** by configuring **AWS IAM** roles and security groups, while integrating stripe **payment gateway**.

Co-Founder | Macverin Technologies | Hybrid (UP, India) | Jul 2022 - Dec 2022

- Led technical strategy and **system design**, architecting scalable distributed solutions using versatile backend stack (**Java Spring, Python Django/FastAPI, Node.js**) automating complex business workflows.
- Engineered **full-stack** architectures using JavaScript ecosystem (**TypeScript, React, Next.js, Express**), creating modular, type-safe interfaces backed by robust APIs to handle data-intensive workloads.
- Directed end-to-end product delivery and DevOps, managing complete SDLC from **UI/UX design** (Figma) to cloud deployment (**AWS, Docker**), and establishing automated **CI/CD pipelines** that ensured 99.9% system availability.

Projects

Wort – Personal Research Assistant | www.wort.nyc | In-Progress

- Developed autonomous research agent using **custom tools and langgraph with RAG pipeline**, implementing a **novel solution-tree algorithm** for multi-step reasoning.
- Developing hybrid system (REST and WebSocket) with State Management, Data Persistence and Caching using **Python, Typescript**.
- Architecting a scalable-dockerized system to be hosted over **AWS** powered by CI/CD.

Snap2Caption - ML Systems for Caption Generation | [code](#) | 2025

- Built a complete **ML pipeline** to generate Instagram-ready captions and hashtags from photos in under 2 seconds.
- **Fine-tuned LLaVA-1.5/1.6 (7B) vision models using LoRA** on 100k urban images for efficient, high-quality training.
- Engineered a **production setup handling 300+ requests/hour**, fully monitored with **MLflow, Prometheus**, and **Grafana**.
- Automated infrastructure provisioning on GPU clusters using **Terraform**.

Finassistant - RAG-based Financial Agent | [live demo](#) | [code](#) | 2025

- Created a conversational financial analysis tool that pulls real-time data from equity, crypto, and macro-APIs.
- Utilizes RAG to search financial news and answer portfolio/risk questions in natural language.
- Currently fine-tuning on sentiment-labeled datasets to "read between the lines" of earnings calls.